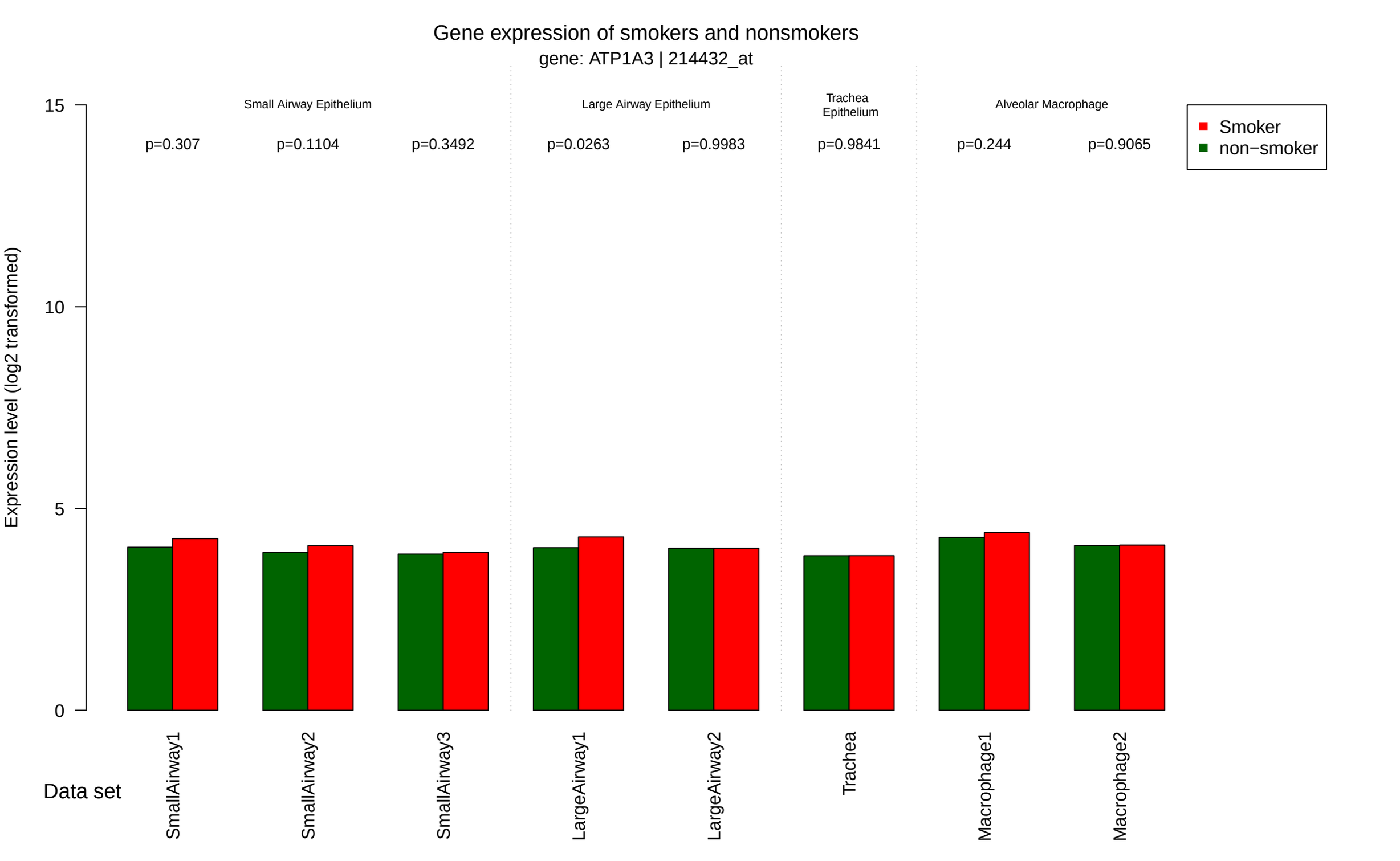
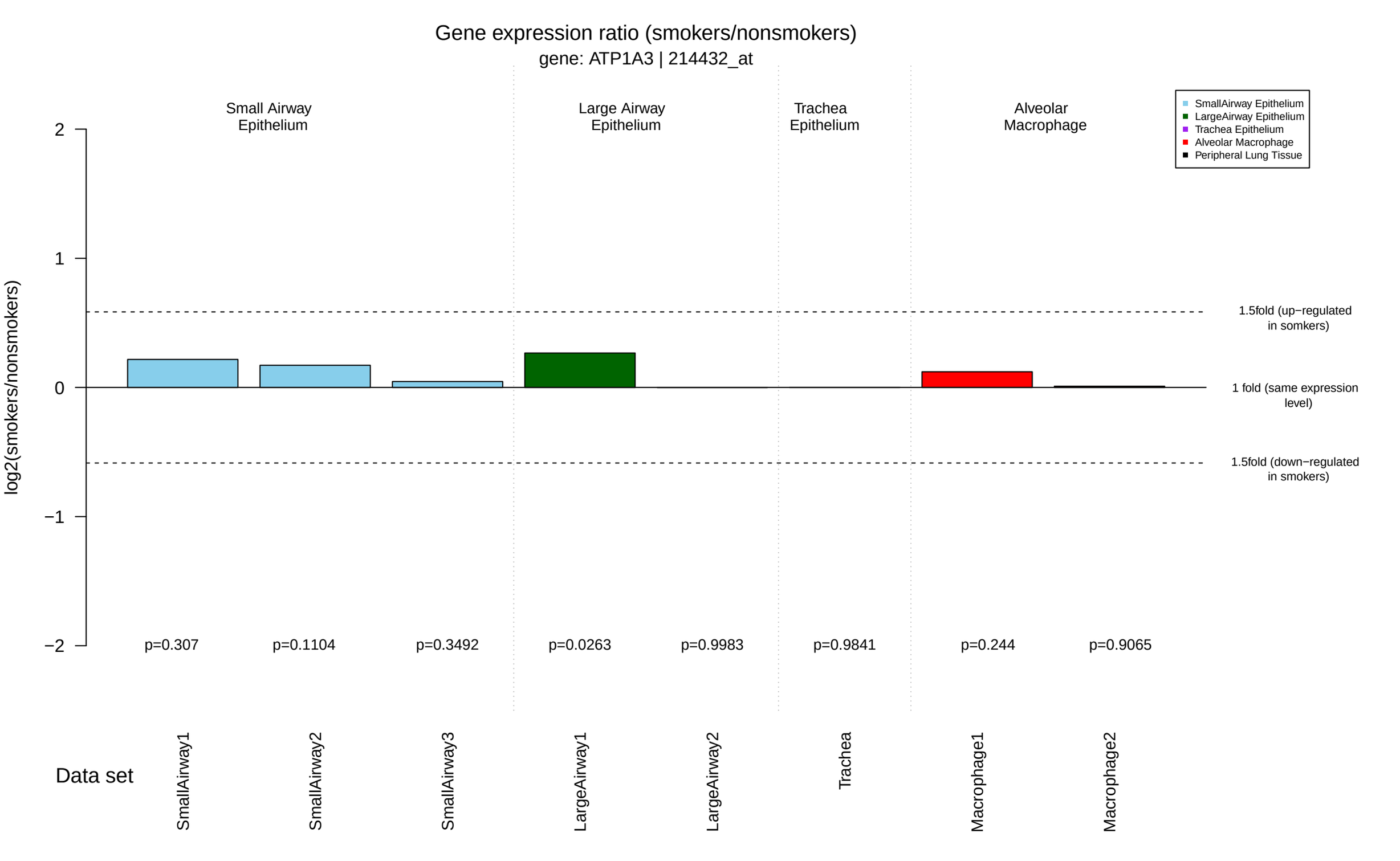
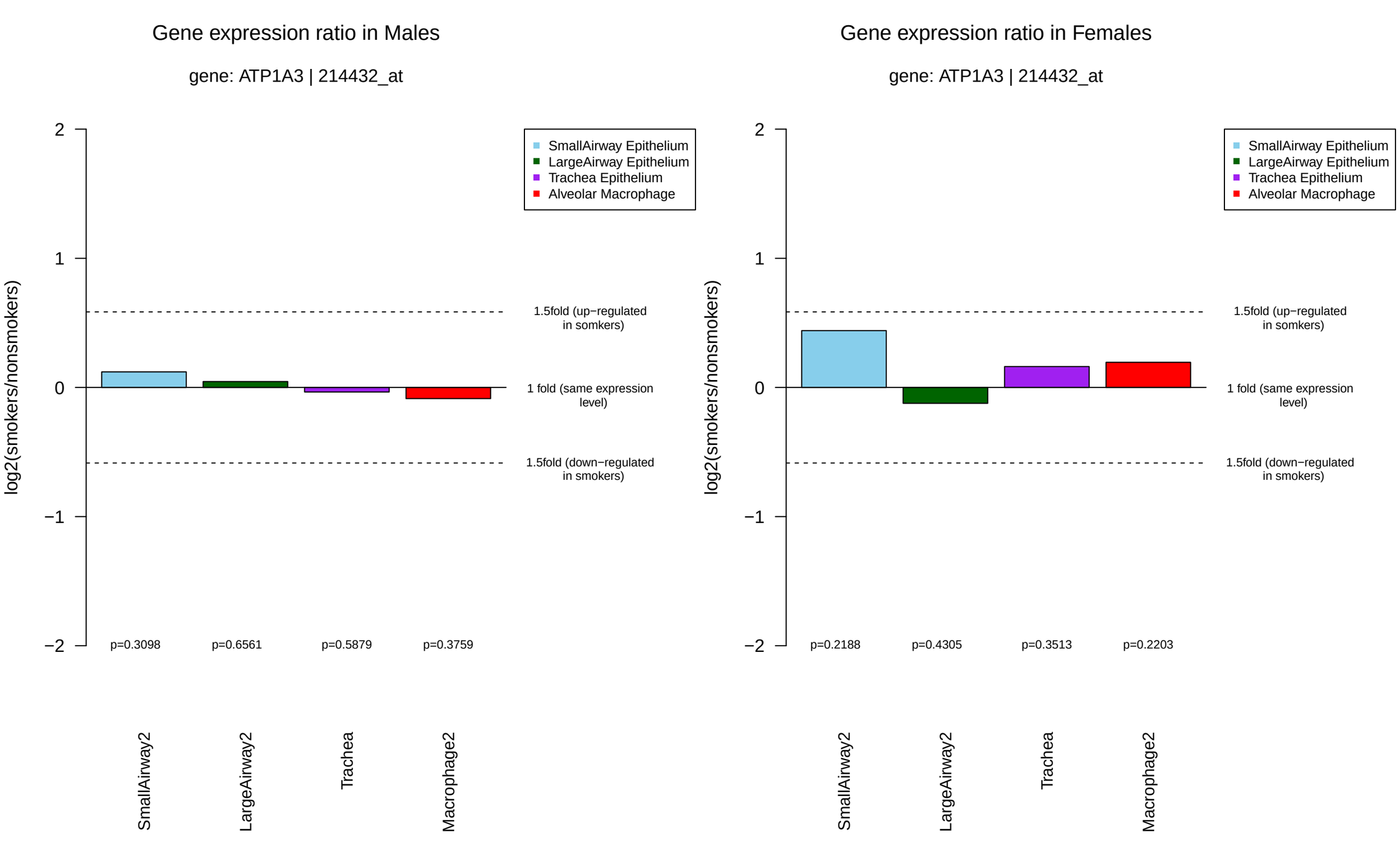
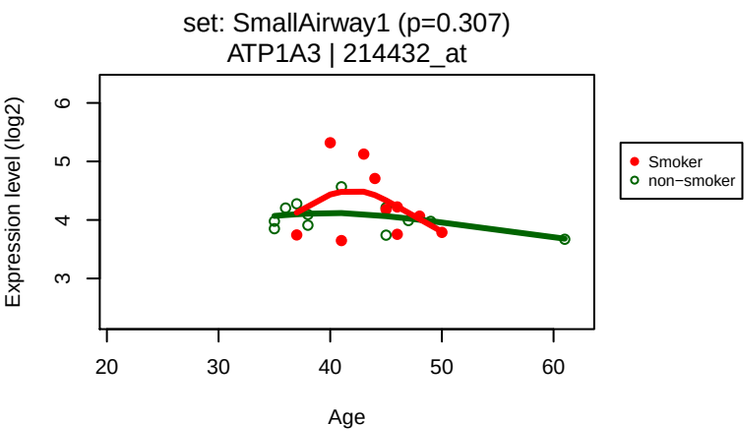
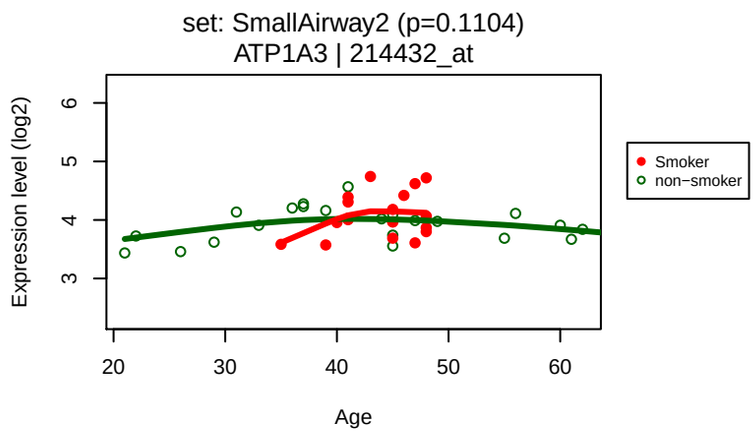
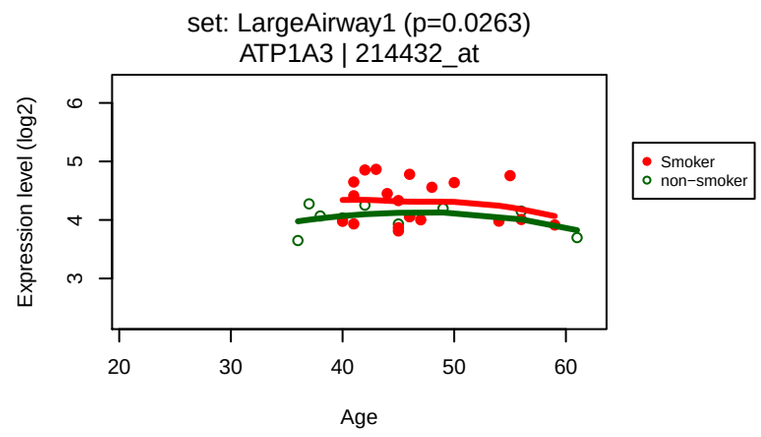

Gene expression of smokers and nonsmokers
gene: ATP1A3 | 214432_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.307
p=0.1104
p=0.3492
p=0.0263
p=0.9983
p=0.9841
p=0.244
p=0.9065
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

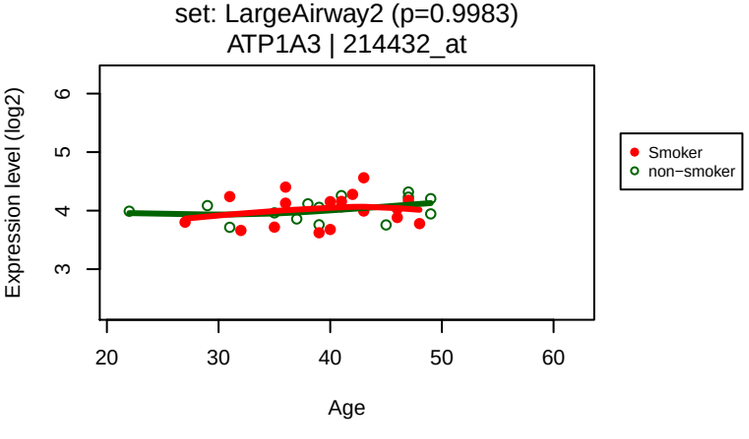

Gene expression ratio (smokers/nonsmokers)
gene: ATP1A3 | 214432_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.307
p=0.1104
p=0.3492
p=0.0263
p=0.9983
p=0.9841
p=0.244
p=0.9065
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

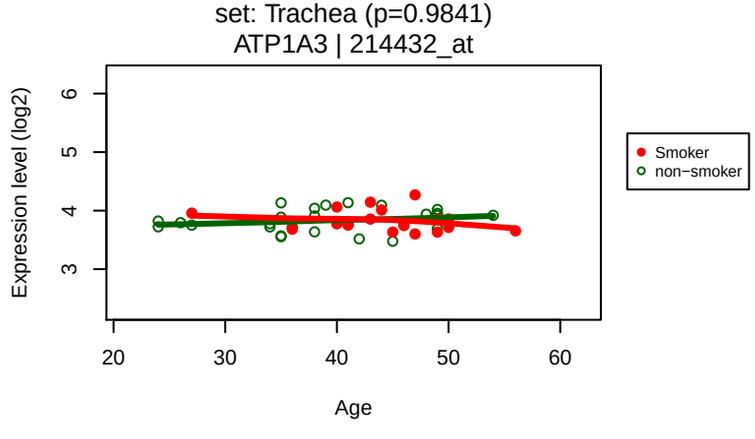

Gene expression ratio in Males
gene: ATP1A3 | 214432_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3098
p=0.6561
p=0.5879
p=0.3759
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP1A3 | 214432_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2188
p=0.4305
p=0.3513
p=0.2203
SmallAirway2
LargeAirway2
Trachea
Macrophage2

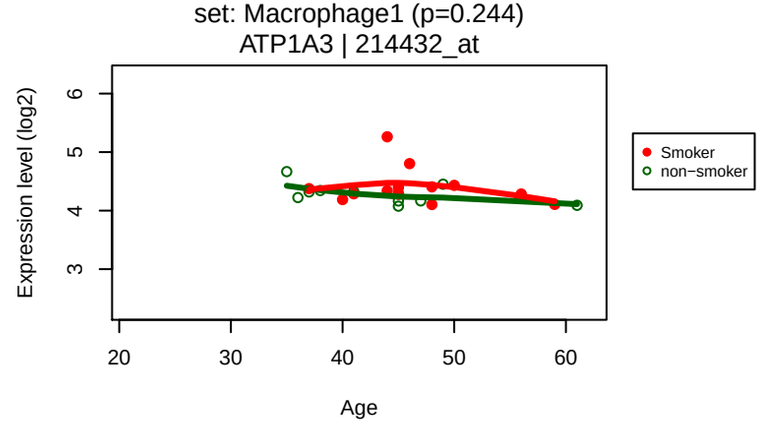

set: SmallAirway1 (p=0.307)
ATP1A3 | 214432_at
Expression level (log2)
Age
Smoker
non−smoker

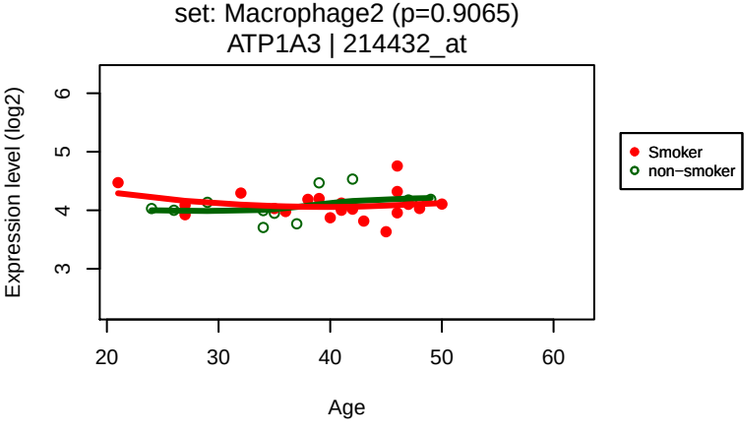

set: SmallAirway2 (p=0.1104)
ATP1A3 | 214432_at
Expression level (log2)
Age
Smoker
non−smoker

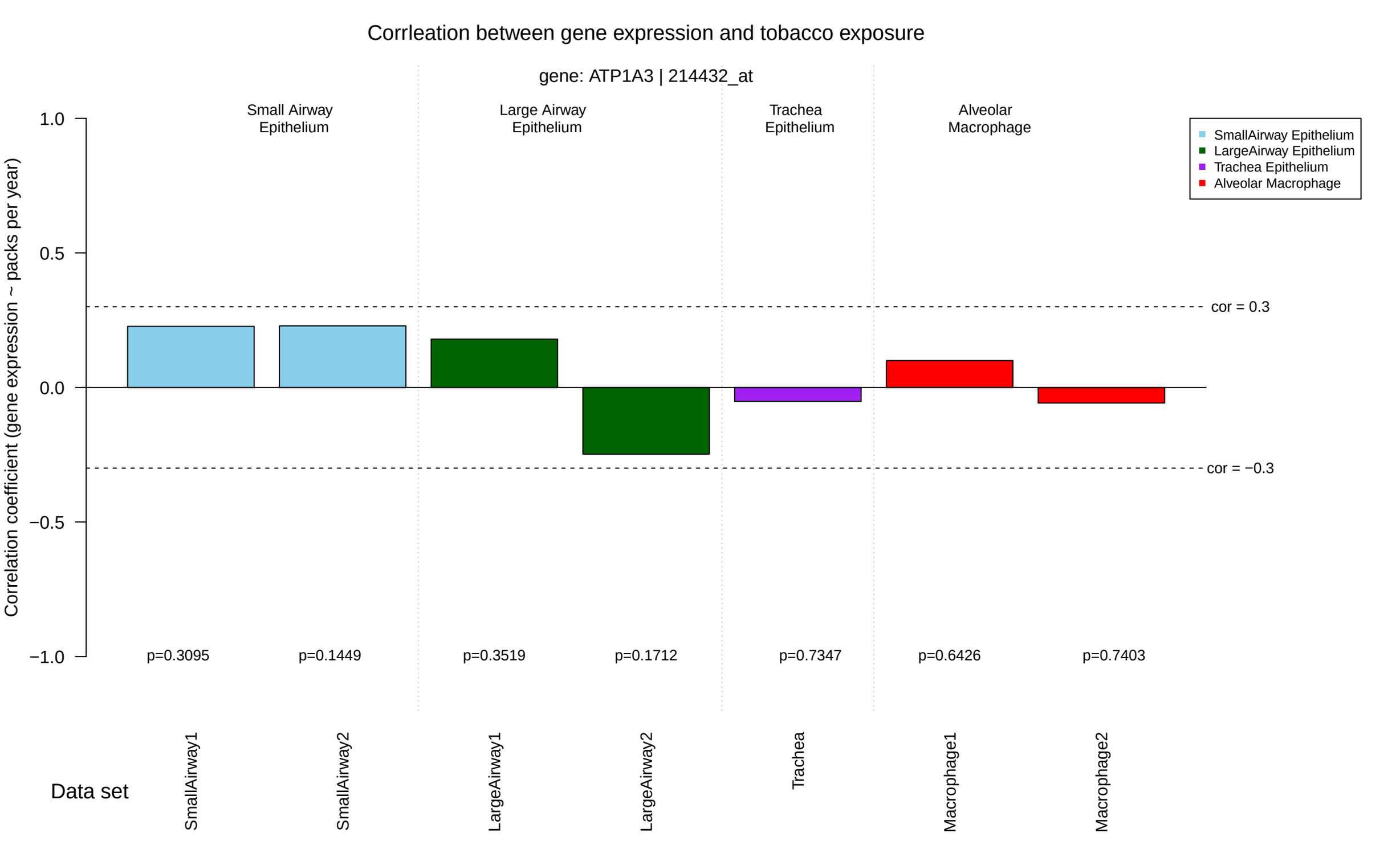

set: LargeAirway1 (p=0.0263)
ATP1A3 | 214432_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9983)
ATP1A3 | 214432_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9841)
ATP1A3 | 214432_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.244)
ATP1A3 | 214432_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9065)
ATP1A3 | 214432_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP1A3 | 214432_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3095
p=0.1449
p=0.3519
p=0.1712
p=0.7347
p=0.6426
p=0.7403
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2